Predictive Analysis of Global Energy Consumption Trends using Time Series and Fine-Tuned LLM Models

Introduction:

The project explores the comparative predictive analysis of Time series and LLM models. Five different models are tested including Statistical Models like ARIMA, SARIMA and LLM Models like Mistral 7b, Prophet (LSTM), TinyLlama 1.1b are fine-tuned on the dataset. All the models are evaluated based on the performance metrics. The main objective is to highlight the comparative analysis of Time Series and LLM Models.

Objectives:

- To preprocess the dataset
- To handle the imbalance in the dataset
- To fine-tune the LLM models
- To evaluate the models using metrics
- To compare all the models
- To provide actionable insights

Explanation of Architecture Diagram:

Energy consumption data is preprocessed into JSONL and CSV format for LLMs and ARIMA, SARIMA models. With LLM trained i.e. fine-tuned iteratively to minimize loss and ARIMA/SARIMA parameters optimized using ADF test. Both models generate forecasts, incorporating seasonal and non-seasonal components. Finally, all the models are evaluated and compared among themselves.

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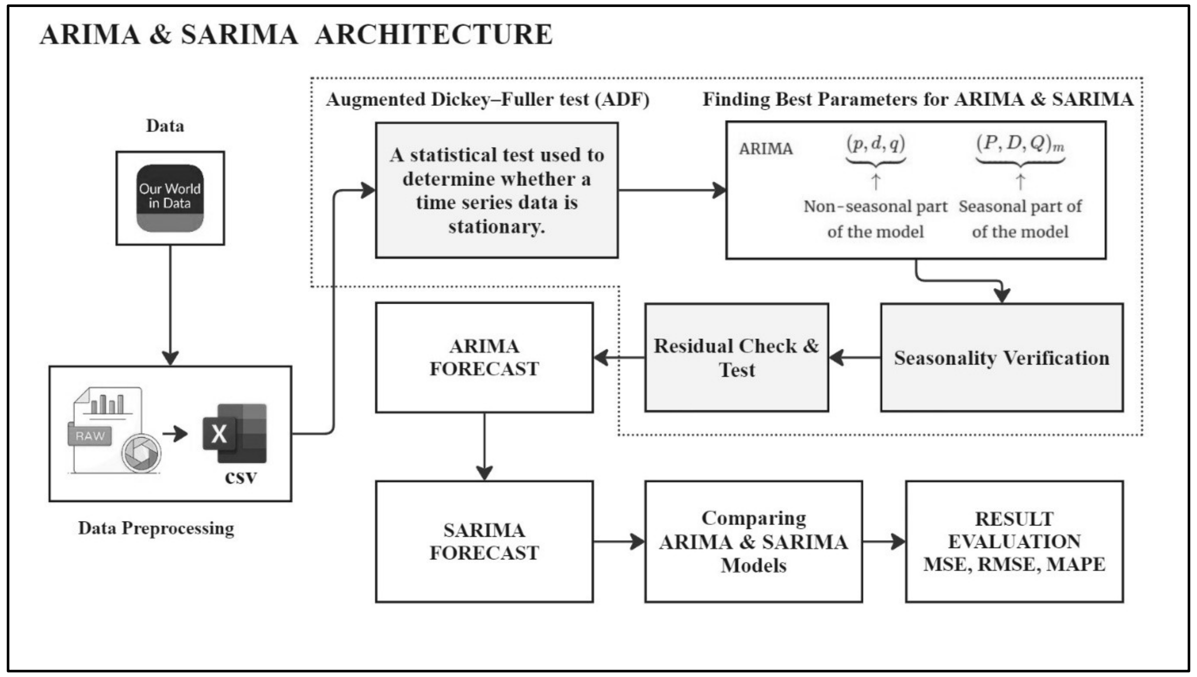
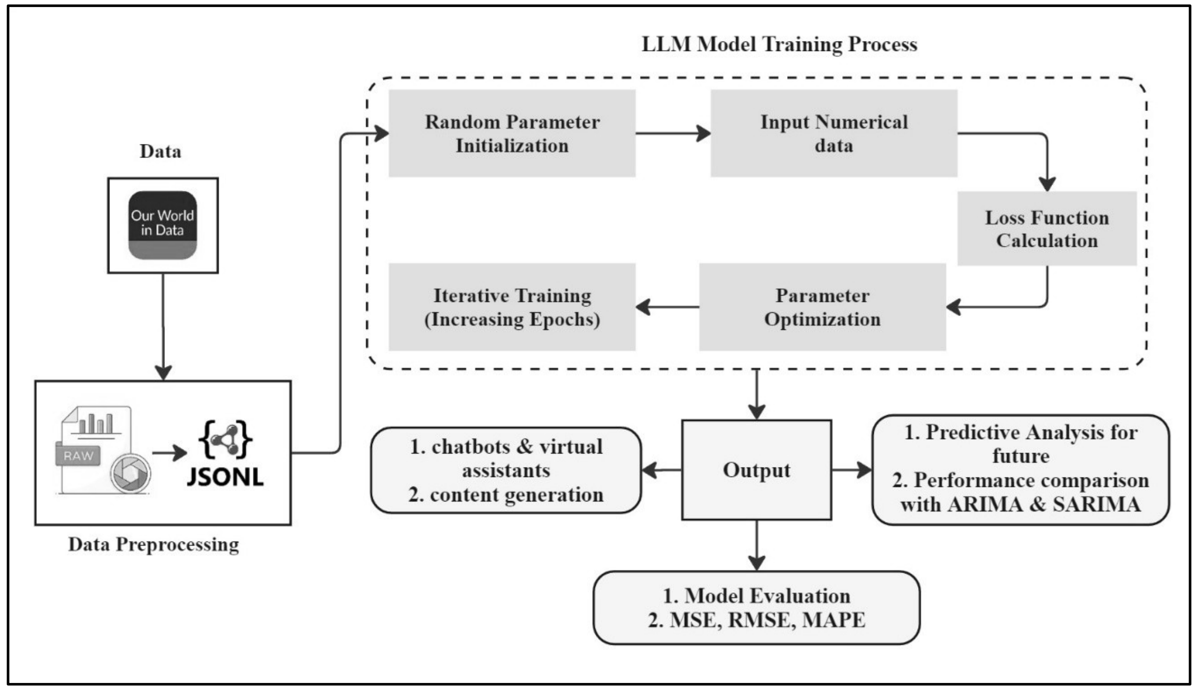
Architecture Diagram:





Conclusion: In conclusion, this project demonstrates the powerful synergy between traditional statistical models and modern AI techniques for analyzing global energy consumption. By combining Time Series models (ARIMA, SARIMA) with Fine-Tuned Large Language Models (LLMs) like Mistral and Llama and LSTM model like Prophet, we were able to capture patterns and trends in the consumption of non-renewable resources such as coal, oil, and natural gas. The results show how AI-driven forecasting can complement classical approaches, offering more adaptive and accurate predictions that are essential for today's rapidly changing energy data. The integration of historical datasets with advanced modeling allows for deeper insights into how energy usage evolves over time and supports the development of smarter, sustainability-focused policies.

Methodology:

- Structurize the dataset into a usable format for processing
- Test Mistral 7b, TinyLlama 1.1b on the dataset
- Use the baseline LLM models & optimize i.e. fine-tune the performance along with iterative training
- Evaluate the models
- Compare the LLM models against the statistical models

Results:

Models	Accuracy	MAE	RMSE	MAPE (%)	R2 Score
ARIMA	0.8677	737.20	846.14	13.23	0.8677
SARIMA	0.9282	398.58	441.95	7.18	0.9082
Prophet LSTM)	0.7963	200.96	240.97	20.37	0.9200
Mistral 7b	0.8223	246.44	386.87	17.74	0.8777
TinyLlama 1.1b	0.9901	4.9250	8.8020	0.91	0.9999